Project Milestone

Graded

Student

Manav Dahra

View or edit group

Total Points

6 / 6 pts

Question 1

Motivation

1 / 1 pt

✔ **– 0 pts** Clear, meaningful problem and motivation.

– 0.5 pts Somewhat clear or lacks depth.

– 1 pt Unclear or missing.

Question 2

Methods

1 / 1 pt

✔ **– 0 pts** Appropriate and well-defined ML approach. (

– 0.5 pts Vague or missing details.

– 1 pt Not described or irrelevant.

Question 3

Preliminary Experiments and Results

2 / 2 pts

✔ **– 0 pts** Clear description of experiments conducted, with early results and meaningful discussion.

– 1 pt Experiments described but incomplete, unclear, or lacking meaningful discussion.

– 2 pts No experiments completed, missing results, or infeasible work.

Question 4

Next Steps

2 / 2 pts

✔ **– 0 pts** Concrete next steps with additional experiments planned and informed by early insights.

– 1 pt Unclear next steps

– 2 pts Missing next steps.

Looks solid!

Geometry-Aware Sharpness Minimization: Invariance and Generalization in SAM Variants

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1 Motivation

Recent advances in deep learning optimization highlight the importance of sharpness-aware methods that seek flat minima to improve generalization. One such method, Sharpness-Aware Minimization (SAM) (Foret et al., 2021), has shown promising results. However, it is sensitive to the choice of model parameterization, as its perturbation radius is defined in the Euclidean metric.

Adaptive SAM (ASAM) (Kwon et al., 2021) and Monge SAM (M-SAM) (Jacobsen and Arvanitidis, 2025) try to address these limitations by introducing scale invariance and geometry-aware metrics, respectively. However, existing evaluations of M-SAM are limited to a single fine-tuning task on ResNet-18 and a multi-modal alignment task; a systematic comparison of all three optimizers under controlled reparametrization across multiple architectures has not been done.

Understanding how these methods behave under reparametrization is crucial for developing more robust optimization techniques, and this gap motivates our empirical work on benchmarking SAM, ASAM, and M-SAM on ResNet-18 and ViT-B/32 architectures with a focus on reparametrization invariance and generalization performance.

2 Methods

We implement three optimizers from scratch (SAM, ASAM, M-SAM) and compare them under controlled conditions.

SGD serves as the reference baseline: vanilla stochastic gradient descent with momentum and weight decay, trained with a cosine-annealing learning rate schedule.

SAM (Foret et al., 2021) augments SGD with a two-step update: parameters are first perturbed to the approximate local loss maximum within an ℓ_2 ball of radius ρ, then a standard gradient step is taken at the perturbed point. Because the perturbation is defined in Euclidean space it is *not* invariant to function-preserving rescalings of the weights.

ASAM (Kwon et al., 2021) addresses this by replacing the ℓ_2 norm with an adaptive (parameter-scaled) norm $T_w = \mathrm{diag}(|w| + \eta)$, making the perturbation scale-invariant: if every weight is multiplied by a scalar α, the perturbation is unchanged.

M-SAM (Jacobsen and Arvanitidis, 2025) takes a Riemannian geometry perspective. The Monge metric on the model-function manifold yields a correction factor $1/\sqrt{1 + \|\nabla_w \mathcal{L}(w)\|_2^2}$ applied to the SAM perturbation, shrinking large-gradient perturbations and promoting geometry-aware, reparametrization-robust updates.

Model and Dataset. We use **ResNet-18** (He et al., 2016) adapted for CIFAR-10 (Krizhevsky, 2009): the standard ImageNet stem (7×7 conv, stride 2, max-pool) is replaced with a 3×3 stride-1 conv and an identity layer to retain spatial resolution for 32×32 inputs. The final fully-connected layer maps from 512 to 10 classes. All experiments use a single random seed (seed 42) at this stage; multi-seed evaluation is planned for the final report.

Training Protocol. All models are trained for 200 epochs on CIFAR-10 with:

- Batch size 256, learning rate 0.1, momentum 0.9, weight decay 5×10^{-4};
- Cosine-annealing LR schedule (no warmup);
- Perturbation radius ρ swept over $\{0.005, 0.01, 0.02, 0.03, 0.05\}$ for SAM and M-SAM; a single $\rho = 0.5$ for ASAM (following the original paper's recommended scale).

For both SAM variants two forward–backward passes are performed per iteration.

Same protocol will be applied to ViT-B/32 on CIFAR-10 as a stretch goal, pending resolution of the GELU reparametrization issue.

Evaluation Metrics. We track final test accuracy, training accuracy, cross-entropy loss curves, and the *divergence rate* $\Delta = \mathcal{L}_{\text{test}} - \mathcal{L}_{\text{train}}$ as a proxy for generalization gap. For reparametrization invariance we measure the variance of final test accuracy across scaling factors $\alpha \in \{0.1, 0.5, 1.0, 2.0, 10.0\}$ applied to a function-preserving ReLU-layer rescaling. Sharpness will be estimated via the Hutchinson stochastic trace estimator (Hutchinson, 1989) $\mathrm{tr}(H)/d$ with Rademacher vectors, which avoids forming the full Hessian.

3 Preliminary Experiments and Results

Baseline Training. We have completed the full baseline sweep (200 epochs each) for all four optimizer (SGD, SAM, ASAM, M-SAM) families on ResNet-18/CIFAR-10, producing 12 checkpoint files: SGD (ρ=0), SAM at five ρ values, ASAM at ρ=0.5, and M-SAM at five ρ values. All models reach 100% training accuracy. Tables 1 and 2 report test accuracy, divergence rate, and Hutchinson sharpness for every configuration; loss landscapes are in the Appendix.

Table 1: Full baseline results on ResNet-18/CIFAR-10 (200 epochs, seed 42). Best row per optimizer shown in bold.

Optimizer	ρ	Test Acc (%)	Div. Rate Δ
SGD	0.000	95.46	0.1777
SAM	**0.020**	**95.75**	**0.1494**
SAM	0.005	95.23	0.1784
SAM	0.010	95.31	0.1729
SAM	0.030	95.70	0.1473
SAM	0.050	95.55	0.1407
ASAM	**0.500**	**95.79**	**0.1440**
M-SAM	0.005	95.47	0.1736
M-SAM	0.010	95.50	0.1683
M-SAM	0.020	95.64	0.1559
M-SAM	**0.030**	**95.76**	**0.1451**
M-SAM	0.050	95.75	0.1384

Table 2: Hutchinson $\mathrm{tr}(H)/d$ for all 12 checkpoints. Lower is flatter.

Optimizer	ρ	$\mathrm{tr}(H)/d$
SGD	0.000	6.25×10^{-4}
SAM	0.005	6.35×10^{-4}
SAM	0.010	4.02×10^{-4}
SAM	0.020	4.45×10^{-4}
SAM	0.030	5.16×10^{-4}
SAM	**0.050**	$\mathbf{3.43 \times 10^{-4}}$
ASAM	0.500	3.17×10^{-4}
M-SAM	0.005	6.24×10^{-4}
M-SAM	0.010	4.29×10^{-4}
M-SAM	0.020	4.40×10^{-4}
M-SAM	0.030	2.61×10^{-4}
M-SAM	**0.050**	$\mathbf{1.89 \times 10^{-4}}$

Several observations stand out. First, SAM at ρ=0.005 is *sharper* than SGD (6.35×10^{-4} vs. 6.25×10^{-4}), confirming that an under-sized perturbation radius fails to flatten the loss landscape and can hurt generalization. Second, M-SAM at ρ=0.05 achieves the lowest sharpness across all runs (1.89×10^{-4}, approximately 70% flatter than SGD), consistent with its best divergence rate. Third, ASAM (3.17×10^{-4}) is flatter than SGD and comparable to the best SAM setting, yet achieves the highest raw test accuracy, suggesting that its adaptive normalization improves generalization through a mechanism not fully captured by the scalar Hessian trace. The full 3D loss landscapes for all 12 checkpoints are shown in Figure 1 (Appendix); per-optimizer best-checkpoint landscapes are in Figure 2.

Infrastructure and Implementation Validation. The training pipeline, data loading (HuggingFace CIFAR-10 cached locally), optimizer implementations, and checkpoint saving have all been validated. The Hutchinson sharpness estimator and loss landscape visualizer are also implemented and unit-tested.

The reparametrization utility (`apply_relu_reparam`) for ResNet-18 BasicBlocks has been implemented and verified. It exploits the positive 1-homogeneity of ReLU, i.e. $\mathrm{ReLU}(\alpha x) = \alpha\,\mathrm{ReLU}(x)$ for $\alpha > 0$, to construct a function-preserving weight scaling. Within each BasicBlock the computation chain is:

$$\xrightarrow{\text{conv1}}\xrightarrow{\text{BN1}}\xrightarrow{\times\alpha}\xrightarrow{\text{ReLU}}\xrightarrow{\times\alpha}\xrightarrow{\text{conv2}}\xrightarrow{\times 1/\alpha}$$

The α factor is injected by scaling BN1's affine parameters ($\gamma \leftarrow \alpha\gamma$, $\beta \leftarrow \alpha\beta$) rather than conv1's weights directly, because BN's normalisation step would otherwise absorb any raw weight scaling. The compensating $1/\alpha$ is applied to conv2's input-channel dimension. Since ReLU is 1-homogeneous, the net effect on conv2's input cancels and the block's output is unchanged for any $\alpha > 0$. `verify_reparam_resnet` confirms a max absolute output deviation of $< 10^{-5}$ across all tested α values.

ViT-B/32 experiments are currently paused: GELU is not exactly 1-homogeneous, so the reparametrization via simple weight scaling introduces a measurable output deviation (~0.06 per unit for $\alpha = 2$). This issue is being investigated; a corrected formulation will be incorporated before the final report.

4 Next Steps

1. **Reparametrization invariance experiment (Experiment 2).** Run the full sweep: for each of the four optimizers, apply the five scaling factors $\alpha \in \{0.1, 0.5, 1.0, 2.0, 10.0\}$ to the ResNet-18 initialization and train for 200 epochs. Report per-optimizer variance of final test accuracy across α values and test the hypothesis that SAM is most sensitive, ASAM is partially invariant, and M-SAM is most invariant.
2. **Multi-seed evaluation.** Re-run the best configurations (SAM ρ=0.02, ASAM ρ=0.5, M-SAM ρ=0.05, SGD) with seeds $\{0, 1, 2\}$ (in addition to seed 42) to obtain SEM estimates and statistically sound comparisons. Current single-seed results show consistent trends but cannot support significance claims.
3. **Sharpness–generalization correlation.** With multi-seed data, compute the Pearson correlation between $\mathrm{tr}(H)/d$ and divergence rate Δ across all optimizer–ρ combinations to quantify how well local sharpness predicts generalization.
4. **ViT experiments (stretch goal).** Resolve the GELU reparametrization approximation and replicate the key experiments on ViT-B/32 to test whether geometry-aware invariance generalises across architectures.

5 Team Contributions

- **Manav Dahra**: End-to-end implementation of all four optimizers (SGD, SAM, ASAM, M-SAM); ResNet-18 and ViT-B/32 model adapters; CIFAR-10 data pipeline; training loop with two-step SAM closure support; reparametrization utility for ResNet-18 BasicBlocks; Hutchinson trace estimator and loss landscape analysis; experiment runner scripts and YAML configuration system; baseline training sweep (12 checkpoint runs); initial results collection and this milestone writeup.

References

Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. 2021. Sharpness-Aware Minimization for Efficiently Improving Generalization. In *International Conference on Learning Representations (ICLR)*. `https://arxiv.org/abs/2010.01412`

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. *CVPR* (2016). `https://arxiv.org/abs/1512.03385`

M. F. Hutchinson. 1989. A Stochastic Estimator of the Trace of the Influence Matrix for Laplacian Smoothing Splines. *Communications in Statistics — Simulation and Computation* 18, 3 (1989), 1059–1076.

Albert Kjøller Jacobsen and Georgios Arvanitidis. 2025. Monge SAM: Robust Reparameterization-Invariant Sharpness-Aware Minimization Based on Loss Geometry. arXiv:2502.08448 [cs.LG] `https://arxiv.org/abs/2502.08448`

Alex Krizhevsky. 2009. Learning Multiple Layers of Features from Tiny Images. (2009). `https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf`

Jungmin Kwon, Jeongseop Kim, Hyunseo Park, and In Kwon Choi. 2021. ASAM: Adaptive Sharpness-Aware Minimization for Scale-Invariant Learning of Deep Neural Networks. In *International Conference on Machine Learning (ICML)*. `https://arxiv.org/abs/2102.11600`

A Supplementary Figures

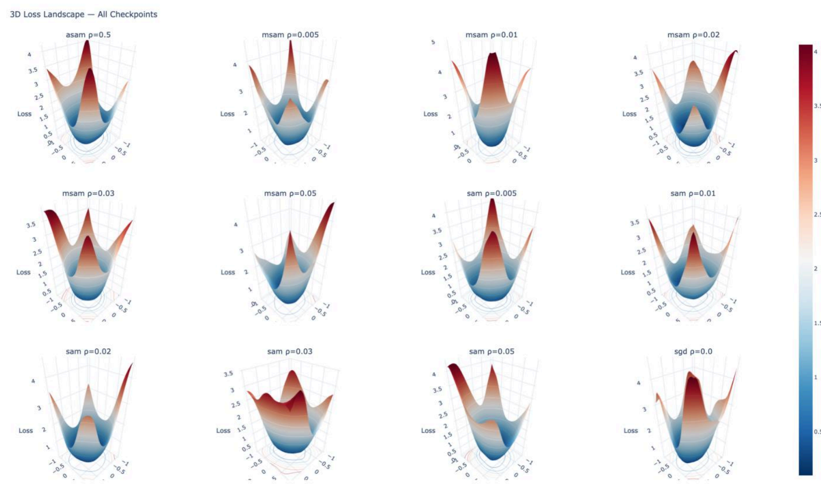
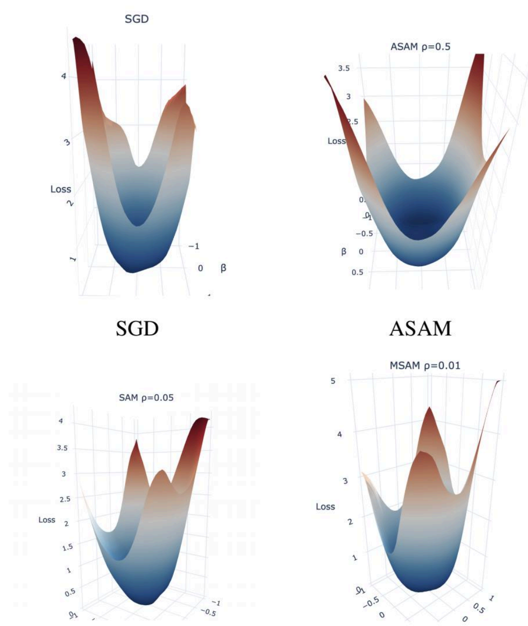


Figure 1: 3D filter-normalized loss landscapes for all 12 checkpoints (SGD, SAM $\times 5$, ASAM, M-SAM $\times 5$) on ResNet-18/CIFAR-10. Grid is 31×31 over $\alpha, \beta \in [-1, 1]$, clipped at loss $= 5.0$. Color encodes loss (blue = low, red = high).



SGD ASAM

SAM (ρ=0.05) M-SAM (ρ=0.05)

Figure 2: 3D filter-normalized loss landscapes for the best checkpoint per optimizer (ResNet-18/CIFAR-10). ASAM and M-SAM show broader, flatter basins than SGD and SAM.

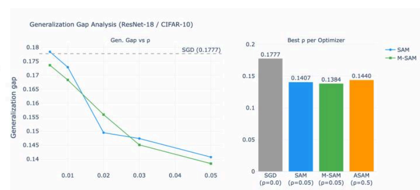


Figure 3: Generalization gap analysis (ResNet-18/CIFAR-10, seed 42). *Top*: divergence rate vs. ρ; M-SAM consistently lower. *Bottom*: best-ρ per optimizer vs. SGD. M-SAM at ρ=0.05 achieves the largest reduction (-22%).